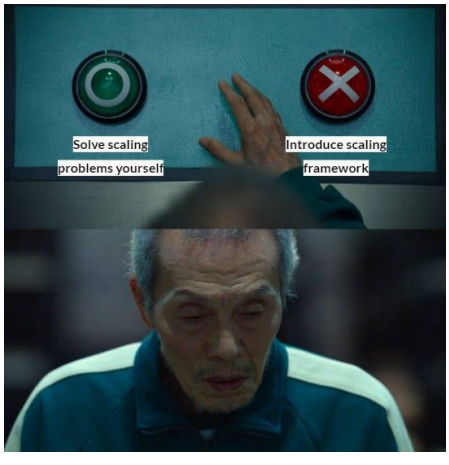
9. Scaling Scrum in Organizations

"Software Development Methodologies"
Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

Motivation





Motivation

- What are the challenges associated into scale Agile?

- Collaboration and Coordination
- Integration and Dependency Management
- Scaling Agile Practices
- Consistency and Alignment
- Communication and Transparency
- Decision-Making and Prioritization
- Scaling Agile Mindset

SAFe (Scaled Agile Framework)

- Scaled Agile Framework® (SAFe®) is a set of organizational and workflow patterns for implementing agile practices at an enterprise scale

- Key Features:
  - SAFe organize work into multiple layers:
    - Team: Agile teams (e.g. Scrum Kanban)
    - Program: multiple teams that are part of an Agile Release Train (ARTs) towards a common goal typically (5-12 teams)
    - Large Solution: enterprise large-scale solutions that go beyond the score of a single ART
    - Portfolio: a set of value streams that delivers a continuous flow of valuable solutions to customers
  - Program Increment (PI): is a timebox during which an Agile Release Train (ART) delivers incremental value in the form of working software and system
  - Lean Portfolio Management (LPM) describes how senior leadership applies lean principles to connect strategy to execution

SAFe (Scaled Agile Framework)

- Scaled Agile Framework® (SAFe®) is a set of organizational and workflow patterns for implementing agile practices at an enterprise scale

- **Best For:** Large enterprises with multiple teams that need formal processes for scaling.

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SAFe (Scaled Agile Framework)

- Scaled Agile Framework® (SAFe®) is a set of organizational and workflow patterns for implementing agile practices at an enterprise scale

- SAFe Principles

#1 Take an economic view

#2 Apply systems thinking

#3 Assume variability; preserve options

#4 Build incrementally with fast, integrated learning cycles

#5 Base milestones on objective evaluation of working systems

#6 Make value flow without interruptions

#7 Apply cadence, synchronize with cross-domain planning

#8 Unlock the intrinsic motivation of knowledge workers

#9 Decentralize decision-making

#10 Organize around value

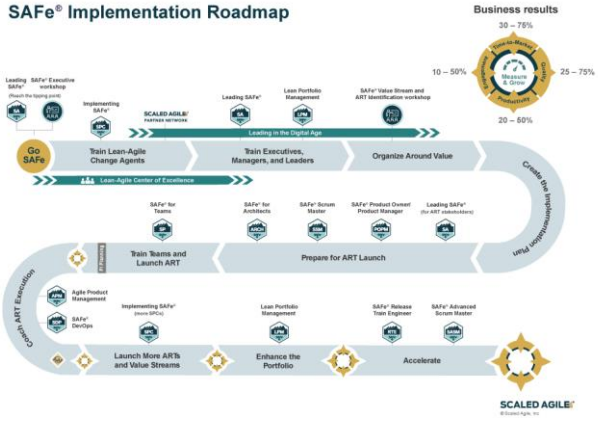
- https://scaledagileframework.com/safe-lean-agile-principles/
- https://www.atlassian.com/agile/agile-at-scale/what-is-safe

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SAFe (Scaled Agile Framework)

- How does SAFe work?
  - executive-level sponsorship
  - Foundation of Scrum



https://scaledagileframework.com/implementation-roadmap/

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LeSS (Large-Scale Scrum)

- framework designed to apply Scrum principles to large-scale product development with multiple teams working together
- all teams share a single Product Backlog and a single Product Owner
- multiple teams (typically 2 to 8 teams) work together on the same product and synchronize their work in a **single Sprint**

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LeSS (Large-Scale Scrum)

- **Sprint at the Same Time**: This ensures alignment and consistency across the product development.

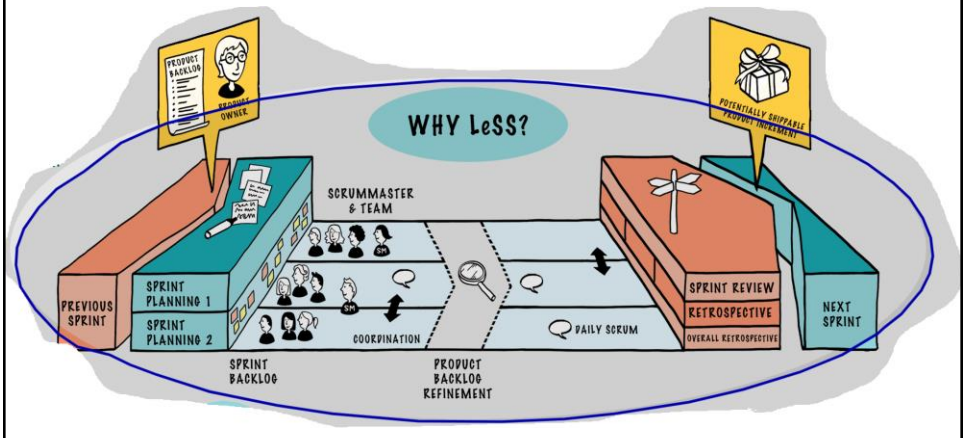
1. **Joint Sprint Planning**: LeSS uses a **two-part Sprint Planning** process:**Sprint Planning 1**: All teams come together to decide what Product Backlog Items (PBIs) they will work on.

1. **Sprint Planning 2**: After the first part, each team conducts its own session to break down the selected PBIs and plan its tasks for the Sprint.

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LeSS (Large-Scale Scrum)



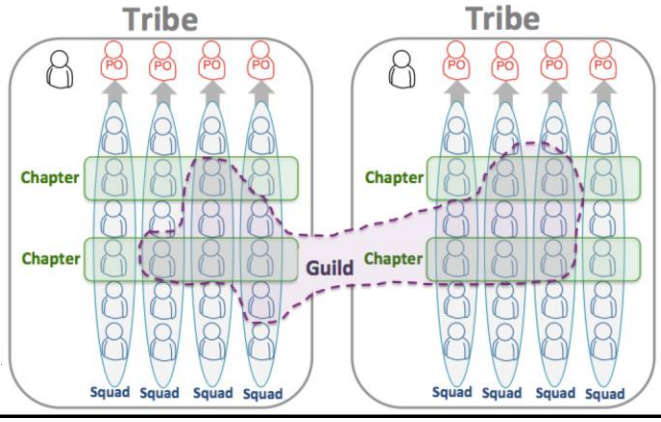
https://less.works/

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Spotify Model

- A technique to scale Scrum up to large groups
- Introduced by Spotify in 2011
- Focus on autonomy of each Squad
  - Good or bad?



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Spotify Model

- Squad: similar to a Scrum team. Self-Organizing team (using Scrum, Kanban, etc) to design, develop, test and release to production

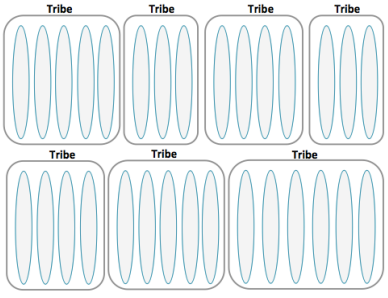


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Spotify Model

- Tribe: *tribe* is a collection of squads that work in related areas – such as the music player, or backend infraestruture.

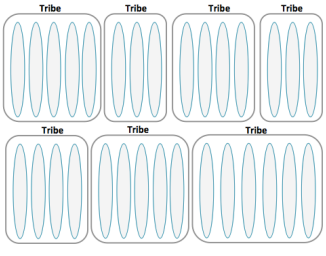


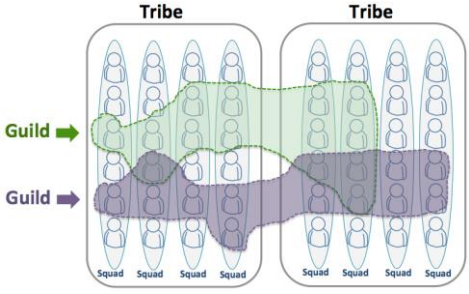
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Spotify Model

- Chapter: group of people having similar skills and working within the same general competency area, within the same tribe
- **Guild:** a "community of interest", a group of people that want to share knowledge, tools, code, and practices



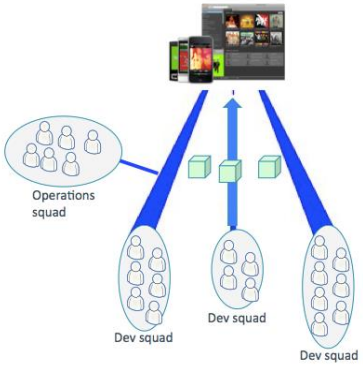


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Spotify Model

- Also an Operation Squad: their job is not to make releases for the squads - their job is to give the squads the support they need to release code themselves; support in the form of infrastructure, scripts, anc routines.

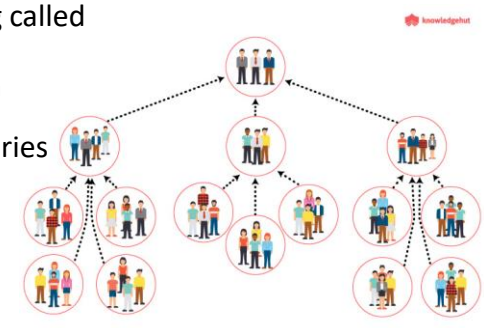


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Scrum@Scale / Scrum of Scrums

- A technique to scale Scrum up to large groups
- Groups are divided into teams of 5-10
- Each team designates one (or more) member - an "Ambassador"
  - Depending on the theme, teams can designate different people
- The Ambassadors from all the teams participate in a daily meeting called the Scrum of Scrums
- The Scrum of Scrum focus on impediments, challenges, solutions, interfaces, boundaries etc. between the teams
  - It has a backlog of its own, where each item contributes to improving between-team coordination



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Nexus

- A framework for scaling Agile, specifically development using Scrum
- Lightweight approach to scaling Scrum
- Limited Applicability (depends of Scrum)

- Key aspects:

  - Nexus Integration Team: concept of a Nexus Integration Team, which consists of representatives from each Scrum team working on the product.
    - Ensure:
      - collaboration, integration, and coordination among the individual Scrum teams

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Nexus

- Key aspects:

  - Nexus Sprint Planning: an event where the Nexus Integration Team, along with the Product Owner, collectively plans and coordinates the work for the upcoming Sprint.

  - Nexus Daily Scrum: synchronization meeting that occurs once per day and involves representatives from each Scrum team.

  - Nexus Sprint Review: event where all Scrum teams, stakeholders, and the Product Owner come together to review the integrated increment of the product.

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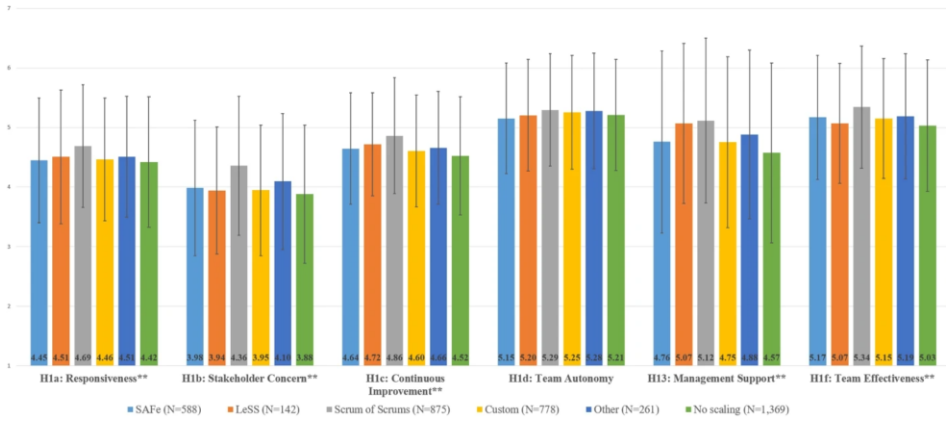
Nexus

- Key aspects:
  - Nexus Sprint Retrospective: a session held at the end of each Sprint where the Nexus Integration Team reflects on the collaboration, integration, and scaling aspects
  - Cross-Team Refinement: cross-team refinement sessions to collaboratively refine and prioritize the Product Backlog
  - Transparent and Integrated Backlog: use of a single, transparent, and integrated Product Backlog that reflects the collective work of all Scrum teams

Nexus

- Key aspects:
  - Nexus Daily Nexus Integration Build: use of a Daily Nexus Integration Build, where teams integrate their work frequently
  - Nexus Definition of Done: establishment of a shared Definition of Done (DoD)
  - Nexus+ Roles: additional roles beyond the traditional Scrum roles to support scaling: Nexus Integration Team member and the Nexus Scrum Master

From: Do Agile scaling approaches make a difference? an empirical comparison of team effectiveness across popular scaling approaches



- (…) we feel that the evidence-based recommendation is for organizations to use the scaling approach that works for them and does not create too much of a mismatch between mindset, structure, and processes.

https://link.springer.com/article/10.1007/s10664-024-10481-5#Fig1

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